



Melanoma-Safe Dermatopathology Decision Support with Foundation Models, Multiple Instance Learning, and Retrieval-Augmented Explainability

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Overview

- Gigapixel H&E skin WSIs; 4 classes: Normal/Benign, BCC, SCC, Melanoma.
- BCC & SCC are the most common; melanoma is the rarest but most lethal.
- Diagnostic evidence is local and sparse; missing melanoma is the costliest error.
- Tiling -> frozen encoders -> gated MIL -> SAFE-R safety -> retrieval + heatmaps.
- A calibrated, selective, evidence-producing workflow, not only a classifier.

Key Contributions

- Melanoma-safe selective diagnosis: accept / review / abstain.
- Cost-aware gated inference: cheap-first UNI -> Phikon -> CONCH.
- Algebra-aware retrieval: diagnostic quotient, risk lattice, AAGS / TRLQ.
- Ensemble explainability: consensus / disagreement / contrastive attention.
- Reproducible, auditable evaluation with PDF / JSON export.

End-to-End Pipeline

1

Input WSI

.svs / .tif gigapixel
OpenSlide read

2

Tissue Tiles

mask + sampling
up to 200 tiles

3

Frozen Encoders

UNI Phikon CONCH

4

Gated MIL

attention pooling
tile bag → slide

5

4-class P(y)

Normal

BCC

SCC

Mel

6

SAFE-R

safety gate
conf / margin / pMel

Accept

auto report
high-confidence

Review / Abstain

selective triage
expert needed

Retrieved Cases

similar-case
audit evidence

Heatmaps

attention /
consensus views

PDF / JSON

audit export
reproducible

Selective evidence production:

confident low-risk slides stop early; uncertain or melanoma-risk slides escalate and produce review, retrieval, and heatmap evidence.

Not an autonomous diagnostic product — a calibrated decision-support prototype.

SAFE-R Decision Flow

UNI probabilities

single-model output for one slide

Normal/Benign

0.08

BCC

0.11

SCC (argmax)

0.42

Melanoma

0.39

Naive classifier → "SCC", done.

argmax 0.42, margin only 0.03,
melanoma 0.39 ignored.

confidence 0.42 < 0.70

UNCERTAINTY FLAG

top-1 margin 0.03 < 0.20

LOW-MARGIN FLAG

pred=SCC but P(Melanoma) 0.39 ≥ 0.20

MELANOMA GUARD

UNI vs Phikon / CONCH disagree

DISAGREEMENT FLAG

SAFE-R → Review / Abstain

automatic SCC prediction is blocked; melanoma cannot be silently dropped

Retrieved neighbors

Heatmap

Audit export

Thresholds shown as small labels on the path, not prose.

Cost-Aware Gated Inference

UNI

cheap-first
foundation model

Safe & confident?

conf ≥ 0.70
margin ≥ 0.20
no melanoma guard

YES

Accept - stop early

UNI only - ~1 encoder

NO

Run Phikon

2nd opinion

Run CONCH

if still unsure

Agree & no guard?

ensemble consensus
melanoma risk cleared

yes

Accept

ensemble decision

no

Review / Abstain

melanoma risk remains

Most slides never run the full 3-model ensemble.

Extra encoders run only on uncertain / melanoma-risk cases — encoder cost ratio = 0.34 vs full 3-model, melanoma FN 0.

Gated-Attention MIL

Tissue tiles

attention a

0.10

0.62

0.18

0.10

Frozen encoder

f(x) → h

Gated attention

$$a = \text{softmax}(w \cdot \tanh(Vh) \odot \alpha(Uh))$$

learns which tiles matter,
with no tile labels

z = Σ a h

bag embedding

classifier

Normal

BCC

SCC

Mel

Attention maps are model evidence, not pathologist tumor annotations.

High-attention tiles show where the model looked — useful for audit and heatmaps, not a verified tumor boundary.

Risk-Aware Retrieval

Query WSI

MIL embedding q

SAFE-R risk tier

Low

Moderate

High

Candidate pool by tier

Low

predicted class only

Mod.

predicted = Melanoma

High

all four class pools

Algebra-aware scoring

AAGS: product of evidence components:
embedding clinical risk-lattice attention

TRLQ: quotient shortlist → rerank top-k
→ far fewer comparisons

Top-k retrieved evidence (audit, not diagnosis transfer)

SCC

sim 0.91

same-label neighbor

SCC

sim 0.88

same-label neighbor

Melanoma

sim 0.79

guard-injected for audit

BCC

sim 0.74

different-label context

SCC

sim 0.72

same-label neighbor

Risk-aware evidence search, not brute-force nearest-neighbor lookup.

SAFE-R 112.6 - TRLQ 64.1 comparisons vs full-cosine 317 — melanoma-risk queries still see a melanoma neighbor.

Data & Setup

4-class mapped subset

2,113 WSIs

Normal/Benign

515

BCC

500

SCC

478

Melanoma

620

+ additional OOD-only pool

sources: COBIRA-derived skin + TCGA-SKCM (melanoma-rich)

314 slides

Frozen test bank

318 slides

79

Normal

75

BCC

71

SCC

93

Melanoma

System setup

Frozen encoders - UNI - Phikon - CONCH

Gated-attention MIL

SAFE-R review gate

Heatmaps + retrieval

PDF / JSON audit export

Triage States

ACCEPT

confidence ≥ 0.62
margin ≥ 0.18
no melanoma guard
models agree

auto report issued
high-confidence, low-risk

REVIEW

! low confidence, or
! low top-1 margin, or
! model disagreement

pathologist review
diagnostic uncertainty

ABSTAIN

melanoma guard:
predMel & P(Mel) ≥ 0.20
OOD warning

no closed-set diagnosis
a designed state, not a failure

Clinical Risk Lattice

Why a lattice?

Missing melanoma costs far more than other confusions.
Safety + retrieval

Melanoma

Highest FN cost - rank 3

BCC

malignant - rank 1

SCC

malignant - rank 1

Normal / Benign

risk rank 0

↑

mis-negative cost

Safety logic and retrieval order classes by clinical error cost — a partial order, not a symmetric class id.

Results at a Glance

1.000

Melanoma recall
retained / ensemble set

0

Melanoma FN
zero missed melanoma

0.986

Retained macro F1
safety-abstain set

0.34

Encoder cost ratio
vs full 3-model

0.9673

Best fixed ensemble F1
UNI + Phikon, melanoma FN 0

0.9603

Gated app macro F1
deployable, melanoma FN 0

0.657

Coverage (safety abstain)
accepted set is smaller but safer

Internal, frozen-test results at a selected operating point — not a universal clinical guarantee.

Macro F1 & Melanoma FN

UNI

best single

FN 1

0.954

UNI+Phikon

fixed ensemble

FN 0

0.967

Safety abstain

retained set (cov 0.66)

FN 0

0.986

Gated app

deployable policy

FN 0

0.960

Adding melanoma-aware safety logic drives melanoma FN to 0 while keeping macro F1 competitive.

Cost Reduction

Encoder calls per slide

Risk = encoders involved

Full 3-model

600

Gated app

~205 (0.34x)

melanoma FN 0 preserved

Retrieval comparisons per query

candidate slides scored

Full cosine

317

SAFE-R routed

112.6 (~64.5%)

TRLQ quotient

64.1 (~79.8%)

Cost is cut by routing compute to risk, not by shrinking the models — stronger encoders stay available for melanoma-risk cases.

Conclusions

- Melanoma-aware safety drives melanoma FN to 0; macro F1 stays 0.96-0.99.
- Risk-aware retrieval cuts comparisons up to ~80% vs full cosine.
- Gated inference uses ~0.34x encoder cost of the full 3-model ensemble.
- Framed as decision support: it asks when to predict, escalate, retrieve, or review.

Limitations

- No prospective / multi-center clinical validation.
- OOD signal is preliminary, not a deployment shield.
- Attention != tumor annotation; retrieval != diagnosis transfer.
- Frozen-test numbers, not a universal guarantee.

References

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